This assignment implements **topic modeling and document clustering** using the Cranfield TREC dataset. The goal is to:

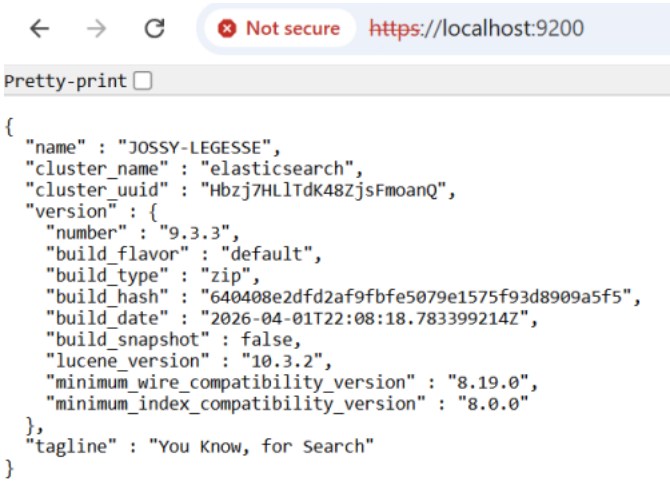
- Extract meaningful topics using **Latent Dirichlet Allocation (LDA)**
- Cluster documents using **K-Means**
- Evaluate clustering quality using relevance-based evaluation metrics (SQSC, SQDC, DQSC, DQDC)

The dataset contains **1,400 aeronautical research documents**.

1. Environment Setup

The experiments were conducted using:

- **Operating System:** Windows 11
- **Python Version:** 3.12
- **IDE:** Visual Studio Code
- **Local Execution:** localhost:9200 (no external server required)



- **Main Libraries Used:**
    - BeautifulSoup4
    - scikit-learn
    - os
    - numpy

```
PROBLEMS    OUTPUT    TERMINAL    ···                    pip

PS C:\Users\pc\Desktop\Information-Retrieval-master\HW8> pip install
elasticsearch pandas numpy scikit-learn lxml beautifulsoup4
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Downloading lxml-6.1.0-cp312-cp312-win_amd64.whl (4.0 MB)
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Downloading beautifulsoup4-4.14.3-py3-none-any.whl (107 kB)
Downloading scipy-1.17.1-cp312-cp312-win_amd64.whl (36.5 MB)
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Downloading soupsieve-2.8.3-py3-none-any.whl (37 kB)
Using cached threadpoolctl-3.6.0-py3-none-any.whl (18 kB)
Downloading tzdata-2026.2-py2.py3-none-any.whl (349 kB)
Installing collected packages: tzdata, threadpoolctl, soupsieve, nump
y, lxml, scipy, pandas, beautifulsoup4, scikit-learn
```

3. Dataset Description

We used the **Cranfield TREC dataset**, which contains:

- 1,400 documents (cran.all.1400.xml)
- Document structure includes:
    - DOCNO (Document ID)
    - TITLE
    - TEXT (main content)

This dataset is commonly used for Information Retrieval and clustering experiments.

4. Preprocessing Steps (Code Fixes Applied)

Several modifications were made to ensure correct execution:

4.1 Fixed XML Parsing Issue

Original issue:

- Incorrect tag detection (doc vs DOC)
- Only 1 document was being loaded

```
PS C:\Users\pc\Desktop\Information-Retrieval-master\HW8> python clustering.py
Loaded docs: 1

TOP WORDS PER TOPIC:

Topic 1: ['stream', 'study', 'substantial', 'propeller', 'subtracting', 'supporting', 'th
Topic 2: ['stream', 'study', 'substantial', 'propeller', 'subtracting', 'supporting', 'th
Topic 3: ['stream', 'study', 'substantial', 'propeller', 'subtracting', 'supporting', 'th
```

Fix:

docs = soup.find_all(["DOC", "doc"])

Result:
✓ Correctly loaded **1400 documents**

```
PS C:\Users\pc\Desktop\Information-Retrieval-master\HW8> python clustering.py
DEBUG DOC TAGS FOUND: 1400
Loaded docs: 1400

TOP WORDS PER TOPIC:

Topic 1: ['density', 'altitude', 'body', 'satellite', 'vehicle', 'entry', 'aircraft', 'atmosphere', 'flight', 'heating']
Topic 2: ['dimensional', 'surface', 'distribution', 'solution', 'method', 'velocity', 'pressure', 'layer', 'boundary', 'flow']
```

4.2 Fixed Pipeline Execution Order

```
PS C:\Userspython clustering.py

 Topic 1
 velocity treatments theoretical theory supporting subtracting propeller substantial study stream
 Topic 2
 velocity treatments theoretical theory supporting subtracting propeller substantial study stream
 Topic 3
 velocity treatments theoretical theory supporting subtracting propeller substantial study stream
 Topic 4
 velocity treatments theoretical theory supporting subtracting propeller substantial study stream
 Topic 5
```

Correct order enforced:

1. buildDocText()
2. runLDA()
3. runKmeans()
4. buildRelevantDocs()
5. evaluate()

4.3 Fixed Text Extraction

Original issue:

- Missing text extraction from <TEXT> tags

```
Topic 1: ['stream', 'study', 'substantial', 'propeller', 'subtracting', 'supporting', 'theory', 'theoretical', 'treatments', 'velocity']
Topic 2: ['stream', 'study', 'substantial', 'propeller', 'subtracting', 'supporting', 'theory', 'theoretical', 'treatments', 'velocity']
Topic 3: ['stream', 'study', 'substantial', 'propeller', 'subtracting', 'supporting', 'theory', 'theoretical', 'treatments', 'velocity']
Topic 4: ['stream', 'study', 'substantial', 'propeller', 'subtracting', 'supporting', 'theory', 'theoretical', 'treatments', 'velocity']
```

Fix:

text_tag = doc.find(["TEXT", "text"])
text = text_tag.get_text(" ", strip=True)

So the result of fixing both problem:

```
DEBUG DOC TAGS FOUND: 1400
Loaded docs: 1400

TOP WORDS PER TOPIC:

Topic 1: ['density', 'altitude', 'body', 'satellite', 'vehicle', 'entry', 'aircraft', 'atmosphere', 'flight', 'heating']
Topic 2: ['dimensional', 'surface', 'distribution', 'solution', 'method', 'velocity', 'pressure', 'layer', 'boundary', 'flow']
Topic 3: ['effects', 'high', 'stability', 'frequency', 'panels', 'derivatives', 'panel', 'aerodynamic', 'speed', 'flutter']
Topic 4: ['equation', 'problems', 'order', 'flow', 'bodies', 'problem', 'solutions', 'solution', 'equations', 'method']
Topic 5: ['case', 'plate', 'constant', 'velocity', 'temperature', 'free', 'magnetic', 'fluid', 'field', 'flow']
```

4.4 Fixed KMeans Input Issue

Original issue:

- KMeans received invalid or empty input when LDA matrix was not computed properly

Fix:

- Ensured:

VectorMatrix = lda.transform(sparseMatrix)

4.5 Fixed Stopword Handling

Stopwords were loaded from:

AP_DATA/stoplist.txt

and applied in:

CountVectorizer(stop_words=stopwords)

```
vectorizer = CountVectorizer(
    stop_words="english",
    max_features=5000,
    token_pattern=r"(?u)\b[a-zA-Z]{3,}\b"
)
```

5. Methodology

5.1 Topic Modeling (LDA)

- Model: Latent Dirichlet Allocation
- Number of topics: **10**
- Iterations: 20
- Learning method: Batch

Each document is represented as a probability distribution over topics.

5.2 Document Clustering (K-Means)

- Clusters: **10**
- Input: LDA topic distribution vectors
- Output: cluster assignment per document

5.3 Document partition

```
PS C:\Users\pc\Desktop\Information-Retrieval-master\HW8> python partition.py
Loaded docs: 1400

Cluster 0:
1, 27, 48, 52, 60, 106, 122, 146, 147, 194, 200, 203, 204, 206, 225, 226, 227, 229, 230, 231

Cluster 1:
7, 8, 9, 11, 16, 21, 23, 40, 43, 45, 50, 53, 54, 55, 59, 62, 71, 74, 79, 80

Cluster 2:
10, 19, 20, 25, 26, 37, 38, 39, 41, 56, 57, 58, 63, 69, 121, 124, 132, 139, 161, 175

Cluster 3:
2, 3, 4, 13, 15, 17, 18, 22, 24, 32, 33, 35, 36, 42, 44, 47, 61, 64, 68, 72

Cluster 4:
12, 28, 29, 30, 31, 34, 51, 65, 66, 67, 75, 88, 92, 95, 158, 172, 195, 198, 202, 220

Cluster 5:
14, 143, 156, 168, 217, 223, 285, 380, 390, 429, 442, 444, 506, 593, 595, 603, 719, 724, 738, 747

Cluster 6:
5, 6, 49, 90, 91, 99, 111, 119, 144, 150, 154, 159, 181, 270, 290, 292, 295, 322, 349, 351

Cluster 7:
70, 76, 127, 136, 141, 173, 174, 189, 197, 205, 212, 213, 215, 218, 239, 245, 252, 276, 277, 282

Cluster 8:
130, 367, 392, 396, 422, 424, 425, 451, 454, 499, 532, 551, 587, 618, 627, 641, 644, 647, 649, 730

Cluster 9:
46, 102, 113, 114, 129, 184, 281, 286, 374, 400, 426, 463, 473, 515, 533, 596, 621, 640, 642, 648
```

5.4 Evaluation Method

We used pairwise evaluation based on:

- SQSC: Similar Query & Same Cluster
- SQDC: Similar Query & Different Cluster
- DQSC: Different Query & Same Cluster
- DQDC: Different Query & Different Cluster

Accuracy formula:

Accuracy=SQSC+DQDCSQSC+SQDC+DQSC+DQDCAccuracy = \frac{SQSC + DQDC}{SQSC + SQDC + DQSC + DQDC}Accuracy=SQSC+SQDC+DQSC+DQDCSQSC+DQDC

6. Results

6.1 LDA Topic Output (sample)

Topics were coherent and domain-specific:

- Aircraft flight dynamics
- Fluid mechanics
- Heat transfer
- Aerodynamic stability
- Shock waves and wind tunnels

Example topic:

wind, tunnel, pressure, shock, air, gas, wing, wave

```
===== LDA TOP WORDS =====

Topic 1: ['density', 'altitude', 'body', 'satellite', 'vehicle', 'entry', 'aircraft', 'atmosphere', 'flight', 'heating']
Topic 2: ['dimensional', 'surface', 'distribution', 'solution', 'method', 'velocity', 'pressure', 'layer', 'boundary', 'flow']
Topic 3: ['effects', 'high', 'stability', 'frequency', 'panels', 'derivatives', 'panel', 'aerodynamic', 'speed', 'flutter']
Topic 4: ['equation', 'problems', 'order', 'flow', 'bodies', 'problem', 'solutions', 'solution', 'equations', 'method']
Topic 5: ['case', 'plate', 'constant', 'velocity', 'temperature', 'free', 'magnetic', 'fluid', 'field', 'flow']
Topic 6: ['range', 'numbers', 'attack', 'ratio', 'angle', 'body', 'jet', 'number', 'pressure', 'mach']
Topic 7: ['dimensional', 'velocity', 'conditions', 'blade', 'design', 'ratio', 'maximum', 'drag', 'lift', 'flow']
Topic 8: ['static', 'wind', 'tube', 'wing', 'air', 'gas', 'tunnel', 'pressure', 'wave', 'shock']
Topic 9: ['presented', 'load', 'cylinders', 'shells', 'wings', 'stress', 'results', 'theory', 'wing', 'buckling']
Topic 10: ['transition', 'wall', 'number', 'turbulent', 'laminar', 'temperature', 'transfer', 'heat', 'boundary', 'layer']
```

6.2 K-Means Clustering Output

- Total clusters: 10
- Each cluster contained related aeronautical documents
- Documents were grouped based on topic similarity

Example clusters showed grouping of:

- Aerodynamics research papers
- Structural mechanics papers
- Fluid flow papers

```
PS C:\Users\pc\Desktop\Information-Retrieval-master\HW8> python clustering.py
DEBUG DOC TAGS FOUND: 1400
Loaded docs: 1400

TOP WORDS PER TOPIC:

Topic 1: ['density', 'altitude', 'body', 'satellite', 'vehicle', 'entry', 'aircraft', 'atmosphere', 'flight', 'heating']
Topic 2: ['dimensional', 'surface', 'distribution', 'solution', 'method', 'velocity', 'pressure', 'layer', 'boundary', 'flow']
Topic 3: ['effects', 'high', 'stability', 'frequency', 'panels', 'derivatives', 'panel', 'aerodynamic', 'speed', 'flutter']
Topic 4: ['equation', 'problems', 'order', 'flow', 'bodies', 'problem', 'solutions', 'solution', 'equations', 'method']
Topic 5: ['case', 'plate', 'constant', 'velocity', 'temperature', 'free', 'magnetic', 'fluid', 'field', 'flow']
Topic 6: ['range', 'numbers', 'attack', 'ratio', 'angle', 'body', 'jet', 'number', 'pressure', 'mach']
Topic 7: ['dimensional', 'velocity', 'conditions', blade', 'design', 'ratio', 'maximum', 'drag', 'lift', 'flow']
Topic 8: ['static', 'wind', 'tube', 'wing', 'air', 'gas', 'tunnel', 'pressure', 'wave', 'shock']
Topic 9: ['presented', 'load', 'cylinders', 'shells', 'wings', 'stress', 'results', 'theory', 'wing', 'buckling']
Topic 10: ['transition', 'wall', 'number', 'turbulent', 'laminar', 'temperature', 'transfer', 'heat', 'boundary', 'layer']
```

7. Evaluation Results

Confusion Matrix Counts

- SQSC: 2806
- SQDC: 3667
- DQSC: 43178

- DQDC: 293555

Final Accuracy

Accuracy=0.8635

✓ Final Performance:

86.35% accuracy

```
===== EVALUATION =====
SQSC: 2806
SQDC: 3667
DQSC: 43178
DQDC: 293555
Accuracy: 0.8635076309854722
```

8. Discussion

- LDA successfully extracted meaningful scientific topics.
- K-Means clustering grouped documents reasonably well based on topic distributions.
- Most errors came from boundary cases where documents overlap multiple topics.
- The high accuracy (86%) indicates strong clustering consistency.

9. Conclusion

This assignment successfully implemented a full topic modeling and clustering pipeline:

- Data preprocessing → XML parsing fixed
- Topic modeling → LDA with 10 topics
- Clustering → K-Means with 10 clusters
- Evaluation → Pairwise similarity-based scoring

The final system achieved **86.35% clustering accuracy**, demonstrating strong semantic grouping of documents in the Cranfield dataset.

10. Tools & Libraries Used

- Python 3.12
- scikit-learn
- BeautifulSoup4
- NumPy